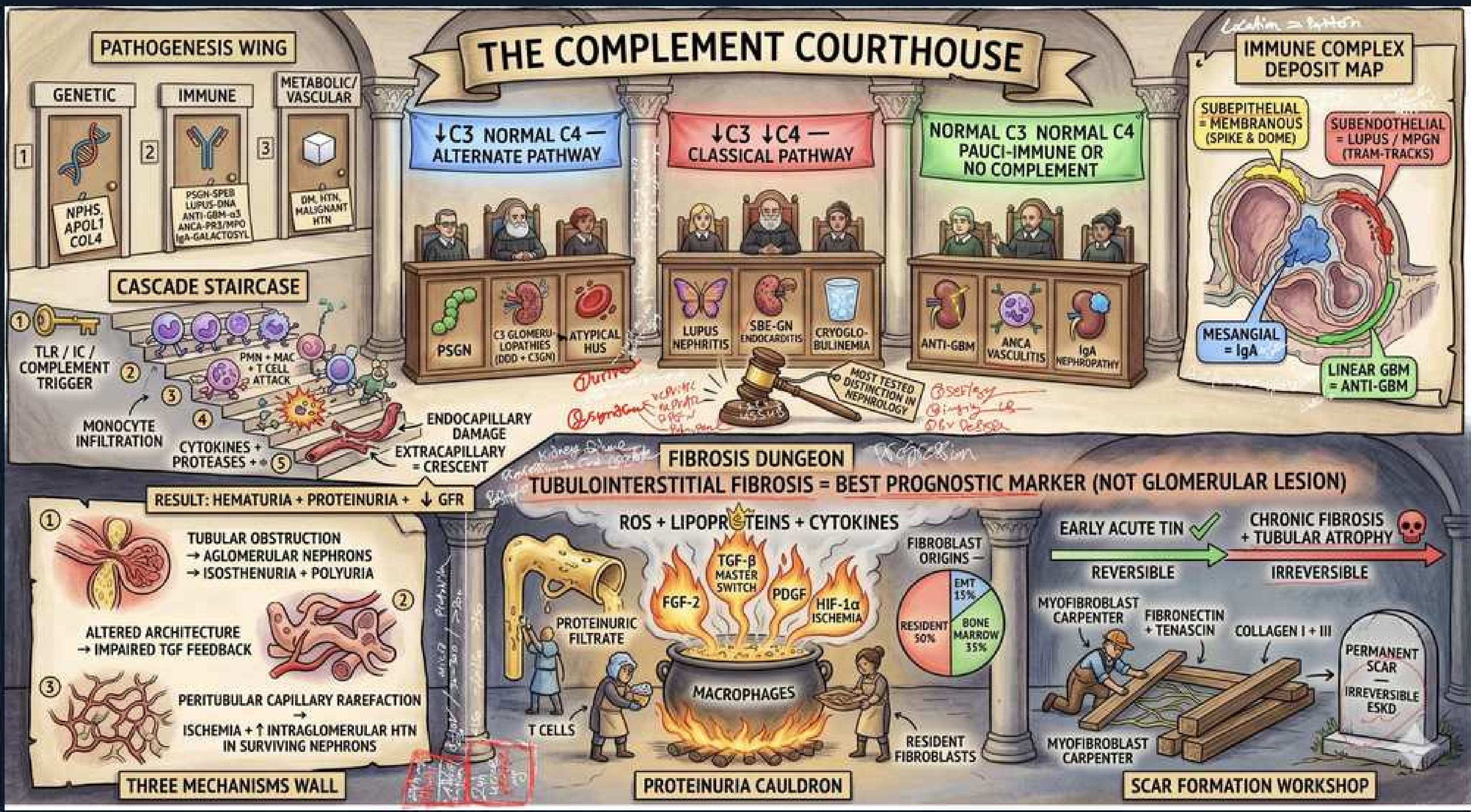
Glomerular Disease — Nephritic Differential by Complement (Complement Courthouse)



1. Low C3, normal C4 — alternate pathway

WHAT YOU SEE

The left judge panel marked with a down-arrow C3 but a normal C4, presiding over PSGN and C3 glomerulopathy — only C3 dropping (alternate pathway bypasses C1/C4) is the visual signature pinning these defendants.

WHAT IT ENCODES

ISOLATED low C3 with NORMAL C4 implicates the ALTERNATE complement pathway: post-streptococcal GN (low C3 that normalizes within ~6–8 weeks) and C3 glomerulopathy (C3 glomerulonephritis and dense deposit disease / 'atypical MPGN', driven by C3 nephritic factor stabilizing the C3 convertase). Because the alternate pathway bypasses C1/C4, C4 stays normal.

BOARD TRAP

WRONG: 'Low C3 with low C4 indicates an alternate-pathway disease like PSGN.' Discriminator — low C3 + low C4 means the CLASSICAL pathway is consumed (lupus, endocarditis, cryoglobulinemia). The alternate-pathway diseases (PSGN, C3 glomerulopathy) drop C3 with PRESERVED C4. The C3/C4 pattern is the key splitter.

2. Low C3 AND low C4 — classical pathway

WHAT YOU SEE

The middle judge panel with BOTH C3 and C4 down-arrows, presiding over lupus, endocarditis, and cryoglobulinemia — both complements consumed (classical pathway via immune complexes) is the bridge to this trio.

WHAT IT ENCODES

Low C3 AND low C4 together signal CLASSICAL-pathway activation by immune complexes: lupus nephritis (low complement tracks activity; anti-dsDNA, full-house IF, C1q), subacute bacterial ENDOCARDITIS-associated GN, and cryoglobulinemic GN (usually HCV-associated; very low C4 with positive rheumatoid factor). Persistently low C3 beyond 8 weeks argues against simple PSGN and toward these.

BOARD TRAP

WRONG: 'PSGN is the prototypical low-C3/low-C4 classical-pathway disease.' Discriminator — PSGN is ALTERNATE pathway: low C3 with NORMAL C4. The low-C3-AND-low-C4 group is lupus, endocarditis, and cryoglobulinemia. If complement stays low past ~8 weeks, reconsider lupus/MPGN rather than resolving PSGN.

3. Normal complement — pauci-immune / anti-GBM

WHAT YOU SEE

The right judge panel showing normal C3 and C4 over ANCA vasculitis and anti-GBM — complement staying normal (no consumption) visually marks these as pauci-immune/anti-GBM rather than immune-complex disease.

WHAT IT ENCODES

NORMAL C3 and C4 point to the pauci-immune and anti-GBM group: ANCA-associated vasculitis (GPA/anti-PR3, MPA/anti-MPO, EGPA — scant immune deposits, 'pauci-immune') and anti-GBM/Goodpasture disease (linear IF, often with pulmonary hemorrhage). IgA nephropathy and HSP also keep complement normal. These are NOT consumptive immune-complex diseases.

BOARD TRAP

WRONG: 'Lupus nephritis has normal complement, so it belongs in this column.' Discriminator — lupus CONSUMES complement (low C3 AND C4), so it sits in the classical-pathway group, NOT here. The normal-complement nephritides are ANCA vasculitis, anti-GBM, IgA, and HSP.

4. Immune-complex deposit map

WHAT YOU SEE

Right wall — EM deposit diagram

WHAT IT ENCODES

EM/IF deposit LOCATION pins the diagnosis: SUBEPITHELIAL humps = PSGN; subepithelial spikes (spike-and-dome) = membranous; SUBENDOTHELIAL deposits with GBM double contours = MPGN / lupus class IV (tram-tracks); MESANGIAL deposits = IgA nephropathy/HSP; LINEAR GBM staining = anti-GBM disease; granular ('lumpy-bumpy') staining = immune-complex disease generally.

BOARD TRAP

WRONG: 'Linear immunofluorescence indicates immune-complex deposition; granular staining indicates anti-GBM disease.' Discriminator — it is the OPPOSITE: LINEAR (smooth) GBM IgG = anti-GBM disease (antibody against a continuous antigen); GRANULAR/lumpy-bumpy = discrete immune-complex deposits (PSGN, lupus, IgA). Do not swap linear and granular.

5. IgA nephropathy — mesangial IgA

WHAT YOU SEE

Mesangial deposit + IgA label

WHAT IT ENCODES

IgA nephropathy (Berger disease) — the most common GN worldwide — shows MESANGIAL IgA (± C3) deposits, NORMAL complement, and classically episodic gross hematuria 1–3 days after a URI/mucosal infection (SYNPHARYNGITIC). Its systemic counterpart is HSP/IgA vasculitis (palpable purpura, arthralgias, abdominal pain). Management: RAAS blockade, with immunosuppression for high-risk/progressive disease.

BOARD TRAP

WRONG: 'Mesangial IgA deposits with hematuria after infection reflect anti-GBM disease.' Discriminator — anti-GBM is LINEAR GBM staining and crescentic, not mesangial IgA. Mesangial IgA with synpharyngitic hematuria and normal complement = IgA nephropathy. Also distinguish timing from PSGN (see next).

6. PSGN — subepithelial humps

WHAT YOU SEE

The cascade staircase showing a strep/immune trigger feeding into the courtrooms with subepithelial 'humps' on the GBM — the camel-hump deposits are the EM fingerprint that points to post-streptococcal GN.

WHAT IT ENCODES

Post-streptococcal GN follows a nephritogenic group-A strep infection — ~1–3 weeks after PHARYNGITIS or ~3–6 weeks after IMPETIGO. EM shows subepithelial 'HUMPS'; complement shows low C3 (alternate pathway) normalizing by ~8 weeks; serology shows rising ASO/anti-DNase B. Children recover well; supportive care (BP/volume control) is the mainstay.

BOARD TRAP

WRONG: 'Hematuria occurring DURING an acute pharyngitis is post-streptococcal GN.' Discriminator — hematuria CONCURRENT with infection (synpharyngitic, 1–3 days) is IgA nephropathy; PSGN is delayed, occurring 1–3 weeks (post-pharyngitis) to several weeks (post-impetigo) AFTER the infection, with low C3. The latency is the discriminator.

7. Endocapillary damage / crescents

WHAT YOU SEE
Center — endocapillary damage, crescent

WHAT IT ENCODES
Inflammatory nephritic injury produces ENDOCAPILLARY proliferation (hypercellular tufts, leukocyte infiltration) and, when severe, CRESCENTS — extracapillary proliferation of parietal epithelial cells, macrophages, and fibrin filling Bowman's space, the histologic signature of RPGN. Crescents on biopsy mandate urgent immunosuppression.

BOARD TRAP
WRONG: 'Diffuse foot-process effacement is the histologic hallmark of crescentic nephritic injury.' Discriminator — foot-process effacement is a PODOCYTE/NEPHROTIC finding (minimal change, FSGS). Nephritic/crescentic injury is defined by endocapillary proliferation and crescents, not effacement.

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8. Result: hematuria + proteinuria + low GFR

WHAT YOU SEE
Left banner — result of glomerular damage

WHAT IT ENCODES
The net NEPHRITIC result of glomerular inflammation is the triad: hematuria (with RBC casts/dysmorphic RBCs), SUB-nephrotic proteinuria (<3.5 g), and declining GFR (oliguria, azotemia, hypertension). This contrasts with the nephrotic tetrad of heavy proteinuria, hypoalbuminemia, edema, and hyperlipidemia.

BOARD TRAP
WRONG: 'Heavy nephrotic-range proteinuria with a bland sediment is the expected result of nephritic glomerular damage.' Discriminator — heavy proteinuria + bland sediment = NEPHROTIC. The nephritic result is an ACTIVE sediment (RBC casts), sub-nephrotic protein, and falling GFR.

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9. Tubulointerstitial fibrosis — best prognostic marker

WHAT YOU SEE
Fibrosis Dungeon banner

WHAT IT ENCODES
Across glomerular diseases, the degree of TUBULOINTERSTITIAL FIBROSIS and tubular atrophy — NOT the glomerular lesion itself — is the single best predictor of long-term renal prognosis, because fibrosis is irreversible and reflects accumulated nephron loss. This principle holds for IgA, lupus, MN, and diabetic kidney disease alike.

BOARD TRAP
WRONG: 'The severity of the glomerular lesion (e.g., crescents, proliferation) is the best long-term prognostic marker.' Discriminator — the best predictor of long-term outcome is the extent of TUBULOINTERSTITIAL FIBROSIS/tubular atrophy, not the glomerular lesion grade; glomerular activity can respond to treatment, but established interstitial scar cannot.

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10. Proteinuria / cytokine cauldron

WHAT YOU SEE
Bubbling cauldron — macrophages, TGF, PDGF

WHAT IT ENCODES
The mechanism linking glomerular injury to interstitial scarring: filtered PROTEINURIA, reactive oxygen species, lipoproteins, and profibrotic CYTOKINES (TGF-β, PDGF) are reabsorbed by/activate proximal tubular cells → recruit macrophages and myofibroblasts → collagen deposition (tubulointerstitial fibrosis). This is why proteinuria reduction (RAAS blockade) is renoprotective.

BOARD TRAP
WRONG: 'Tubulointerstitial fibrosis is driven primarily by complement deposition in the interstitium.' Discriminator — interstitial fibrosis is driven by proteinuria-induced tubular injury and profibrotic mediators (TGF-β, PDGF, ROS), NOT direct complement action. This is the rationale for antiproteinuric therapy to slow progression.

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11. Acute TIN — reversible

WHAT YOU SEE
Right — early acute TIN, reversible scaffolding

WHAT IT ENCODES
EARLY/ACUTE tubulointerstitial injury (edema, inflammatory infiltrate, intact tubular architecture) is potentially REVERSIBLE if the driver is removed and treatment is timely (e.g., stopping a culprit drug, treating the glomerular disease). This is the therapeutic window before permanent scar forms.

BOARD TRAP
WRONG: 'Tubulointerstitial injury is uniformly irreversible regardless of stage.' Discriminator — EARLY/acute interstitial injury is reversible if treated; only the CHRONIC fibrotic/atrophic stage is permanent. Recognizing the reversible window justifies prompt intervention.

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12. Chronic fibrosis / tubular atrophy — irreversible

WHAT YOU SEE
Right — chronic fibrosis, permanent scar

WHAT IT ENCODES
CHRONIC interstitial fibrosis with tubular atrophy represents permanent collagen scarring and nephron dropout — IRREVERSIBLE. Once established, it does not respond to immunosuppression; care shifts to slowing further loss (BP and proteinuria control, RAAS blockade) and CKD management.

BOARD TRAP
WRONG: 'Aggressive immunosuppression can reverse established interstitial fibrosis and tubular atrophy.' Discriminator — immunosuppression may quiet ACTIVE inflammation but CANNOT reverse established fibrosis/atrophy; escalating immunosuppression for chronic scar adds toxicity without benefit. Treat early inflammation; manage chronic scar conservatively.
